

Influences of data processing on the interpretation of atmospheric spectra from JWST

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CONTEXT

- Observational characterisation of exoplanetary atmospheres through **transmission spectroscopy**
- JWST provides unprecedented spectroscopic precision and wavelength coverage
- Inversion techniques (*atmospheric retrieval*) **depend on the underlying data set**

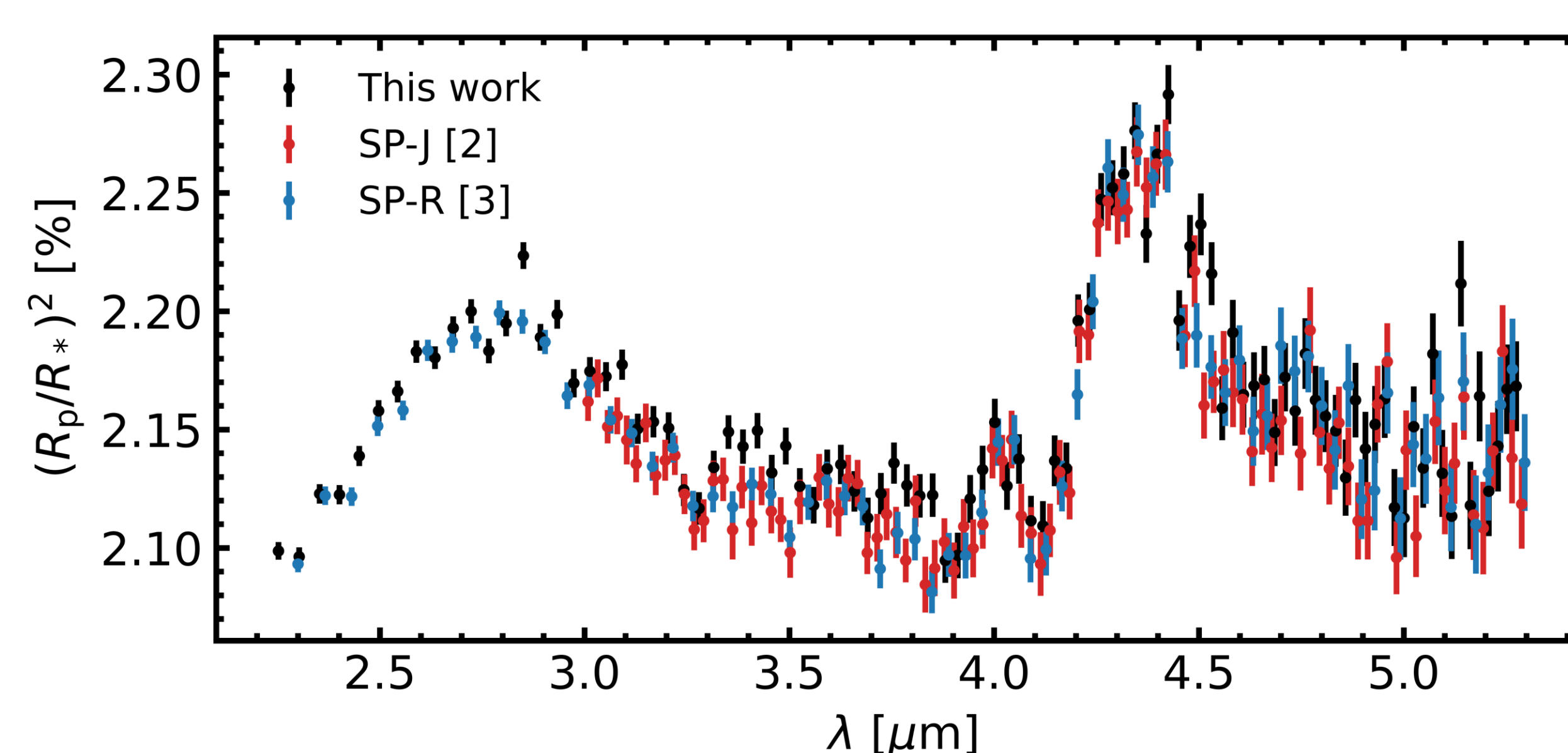
OBJECTIVE

What is the impact of data reduction assumptions on atmospheric characterisation results?

Compare results of **homogenised atmospheric retrievals** on a variation of data reduction results of the same observational data set.

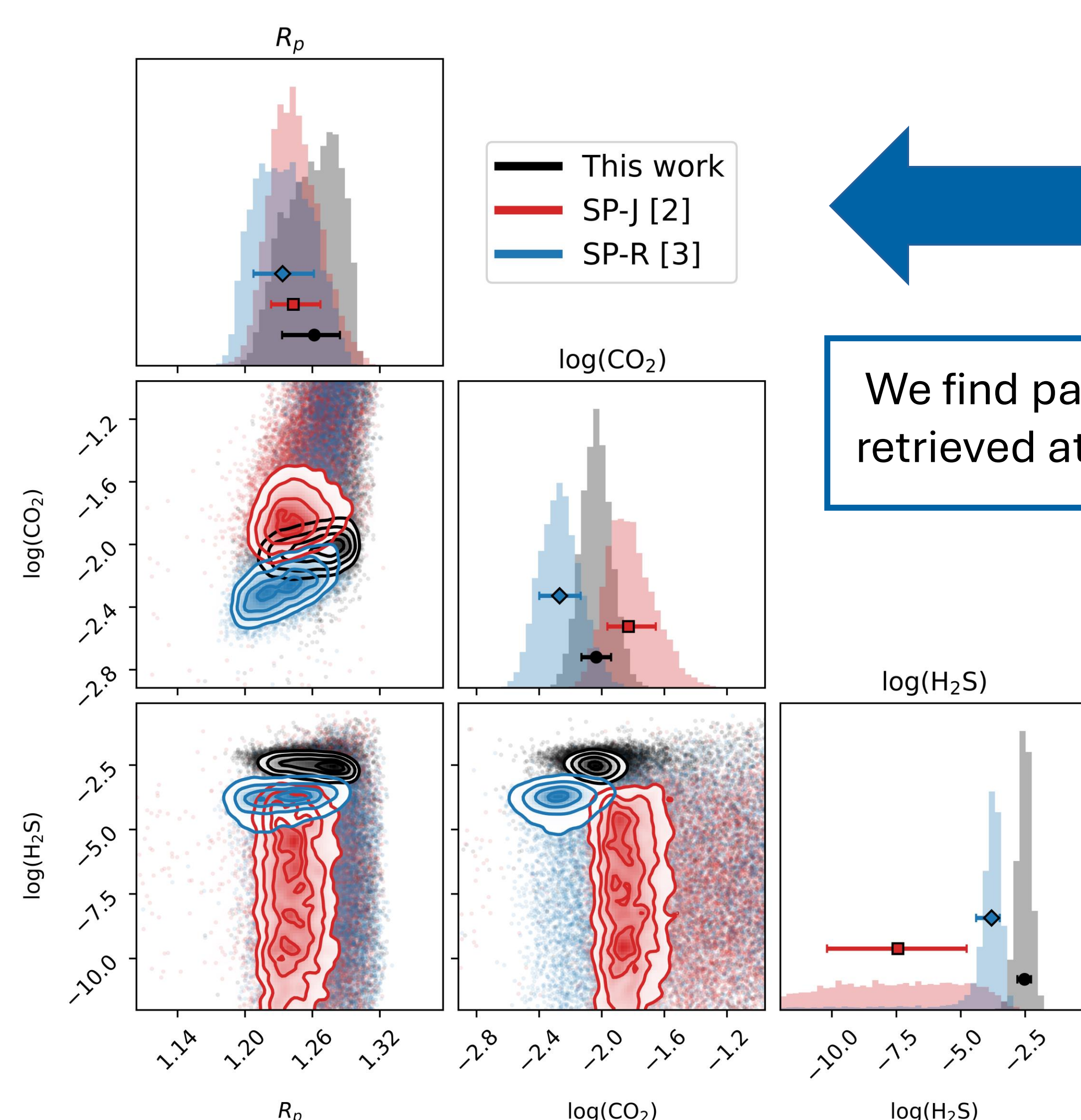
REDUCING THE DATA

- Data reduction of the NIRSpec PRISM observation of the Hot Jupiter **WASP-39 b** (PID: 1366), using *Eureka!* [1]
- Comparison to established data products [2,3] (in red and blue, respectively)

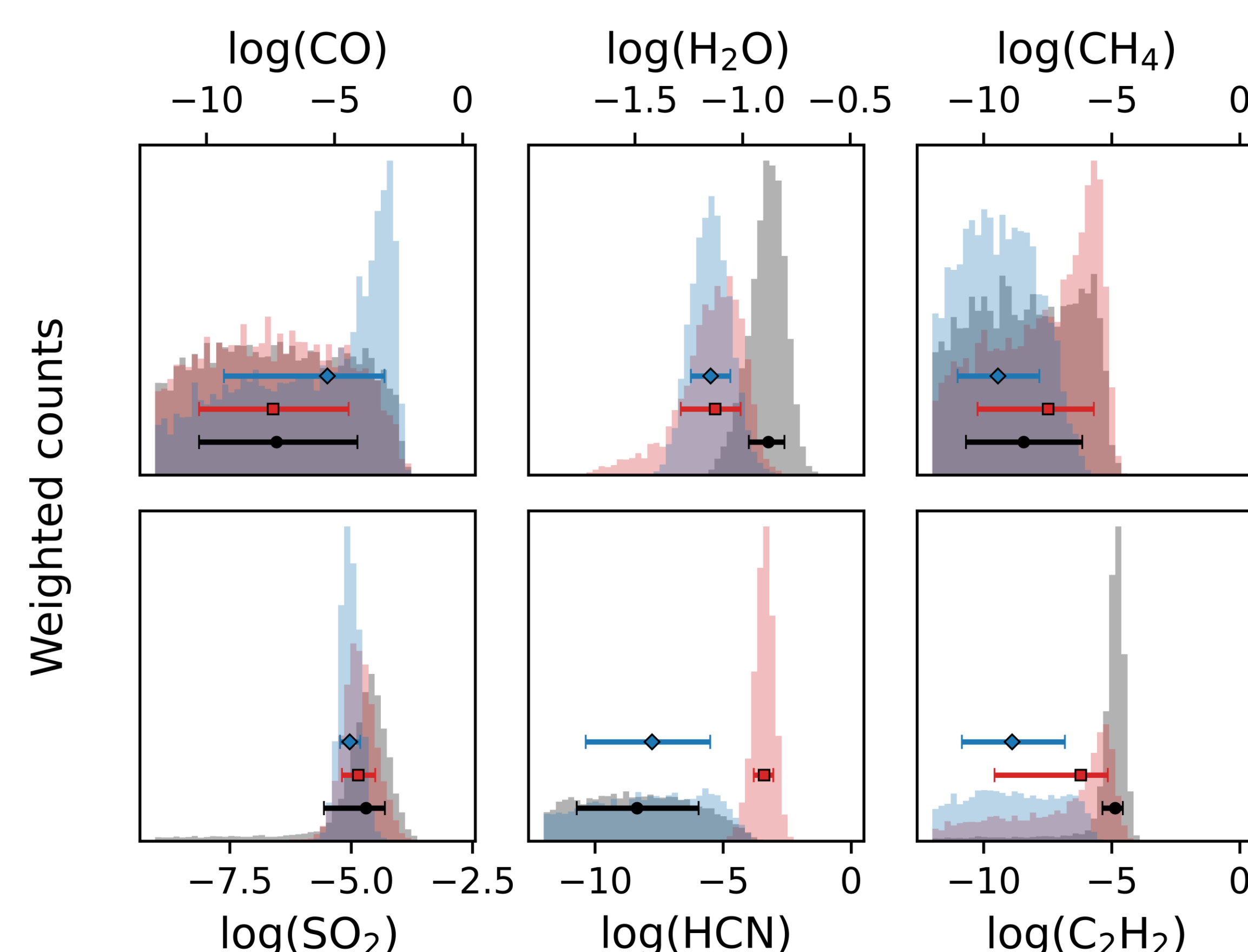


CHARACTERISATION RESULTS

- We perform atmospheric retrieval using **TauREX3** [4]
- Bayesian model selection on the data set produced in this work
- Homogenised atmospheric retrievals on all three considered transmission spectra (*derived from the same observation*)



We find partial agreement in the retrieved atmospheric properties



OUTLOOK

Results of atmospheric characterisation are **sensitive to small-scale variations** in the under-lying, high-precision data.

Retrieval of atmospheric properties will need to account for disagreement based on the influence of underlying data reduction assumptions.

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